* Method of Variation of Parameters :-

This method is applicable to find the G.S. of any linear non homogeneous O.D.E. of second and higher order with constant as well as variable co-efficients.

- Here, the particular integral of the non-homo. eq^{n} is obtained by varying the parameters. i.e. by replacing the arbitrary constants in the C.F. by variable Functions.

- The linear non-homogeneous second order O.D.E. with constant co-efficients is given by

$$\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2 y = Q(x) \quad \text{---(1)}$$

- Let, the C.F. be $C.F. = c_1 y_1(x) + c_2 y_2(x)$ —(2)

According to this method, arbitrary constants c_1 & c_2 in eq^{n} (2) are replaced by two unknown functions $u_1(x)$ & $u_2(x)$ resp. So that

$$P.I = u_1(x)\, y_1(x) + u_2(x)\, y_2(x) \quad \text{---(3)}$$

where $$u_1(x) = -\int \frac{Q(x)\, y_2(x)}{W}\, dx, \quad \text{---(4)}$$

$$u_2(x) = \int \frac{Q(x)\, y_1(x)}{W}\, dx. \quad \text{---(5)}$$

and W = Wronskian of $y_1, y_2 = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

∴ The G.S. of (1) is $y = y_c + y_p$

$$y = c_1 y_1 + c_2 y_2 + u_1 y_1 + u_2 y_2.$$

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* Note:-

1) One may get the same answer no matter which one chosen to be $y_1(x)$ and which one $y_2(x)$ in y_c.

i.e. One may write like $y_c = c_1 y_2(x) + c_2 y_1(x)$.

2) The C.F. is absolutely required to solve the problem.

3) In order to find y_p (or P.I.) given by eq^{n} (3), we will have to solve couple of integrals as given by eq^{n} (4) & (5) and there is no gurantee that we will be able to solve this integrals easily. Sometimes these integrals becomes very difficult particularly in case of variable coefficients.

4) The above method can also be extended to third-order D.E.

- Let, the C.F. of a third-order D.E. be

$$C.F. = c_1 y_1 + c_2 y_2 + c_3 y_3$$

Let, $P.I = u(x) y_1 + v(x) y_2 + w(x) y_3$.

where, $$u(x) = \int \frac{(y_2 y_3' - y_3 y_2')}{W} Q \, dx$$

$$v(x) = \int \frac{(y_3 y_1' - y_1 y_3')}{W} Q \, dx$$

$$w(x) = \int \frac{(y_1 y_2' - y_2 y_1')}{W} Q \, dx$$

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Wronskian $W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$

1) Find the C.F. as ~~C.F. = $C_1 y_1 + C_2 y_2$~~

* Examples:-

1) Solve $y'' + 9y = \sec 3x$

→

The A.E. is $m^2 + 9 = 0$.

$$m = \pm 3i$$

$$C.F. = e^{0x}(C_1 \cos 3x + C_2 \sin 3x)$$

$$y_1 = \cos 3x, \quad y_2 = \sin 3x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{vmatrix} = 3$$

Let, $P.I = u \cos 3x + v \sin 3x$ ——(1)

$$u = -\int \frac{y_2(x)\, Q}{W}\, dx$$

$$= -\int \frac{\sin 3x \sec 3x}{3}\, dx$$

$$= -\frac{1}{3}\int \frac{\sin 3x}{\cos 3x}\, dx$$

$$= -\frac{1}{3}\int \tan 3x\, dx$$

$$= -\frac{1}{3}\left(-\frac{1}{3}\log \cos 3x\right)$$

$$= \frac{1}{9}\log \cos 3x.$$

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$$v = \int \frac{y_1(x)\, Q(x)}{W}\, dx$$

$$= \int \frac{\cos 3x \sec 3x}{3}\, dx$$

$$= \frac{1}{3}\int dx$$

$$= \frac{x}{3}$$

From (1),

$$\text{P.I.} = \frac{1}{9}(\log \cos 3x)\cos 3x + \frac{x}{3}\sin 3x.$$

2) Solve $y'' + a^2 y = \tan ax$.

→

The A.E. is $m^2 + a^2 = 0$

$\therefore m = \pm ai$

$$\text{C.F.} = e^{0x}(C_1 \cos ax + C_2 \sin ax)$$

$y_1 = \cos ax$, $y_2 = \sin ax$

$W = a$

Let, P.I. $= u \cos ax + v \sin ax$. ——— (1)

where, $u = \int -\frac{y_2 Q}{W}\, dx$

$$= \int -\frac{\sin ax \tan ax}{a}\, dx$$

$$= -\frac{1}{a}\int \frac{\sin^2 ax}{\cos ax}\, dx.$$

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$$= -\frac{1}{a}\int \frac{1-\cos^2 ax}{\cos ax}\,dx$$

$$= -\frac{1}{a}\int (\sec ax - \cos ax)\,dx$$

$$= -\frac{1}{a}\cdot\frac{1}{a}\log(\sec ax + \tan ax) + \frac{1}{a^2}\sin ax$$

$$= \frac{1}{a^2}\sin ax - \frac{1}{a^2}\log(\sec ax + \tan ax).$$

and $v = \int \frac{y_1 Q}{W}\,dx$

$$= \int \frac{\cos ax \cdot \tan ax}{a}\,dx$$

$$= \frac{1}{a}\int \sin ax\,dx$$

$$= \frac{1}{a}\left(\frac{-\cos ax}{a}\right)$$

$$= -\frac{1}{a^2}\cos ax$$

From (1),

$$P.I = \left\{\frac{1}{a^2}\sin ax - \frac{1}{a^2}\log(\sec ax + \tan ax)\right\}\cos ax$$

$$-\frac{1}{a^2}\cos ax \cdot \sin ax$$

$$= \frac{1}{a^2}\sin ax\cos ax - \frac{1}{a^2}\cos ax \cdot \log(\sec ax + \tan ax)$$

$$-\frac{1}{a^2}\cos ax \cdot \sin ax$$

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$$P.I. = -\frac{1}{a^2} \cos ax \cdot \log(\sec ax + \tan ax)$$

Hence, the G.S. is

$$y = C_1 \cos ax + C_2 \sin ax - \frac{1}{a^2} \cos ax \cdot \log(\sec ax + \tan ax)$$

(3) Solve $y'' - 3y' + 2y = e^x$.

→

The A.E. is $m^2 - 3m + 2 = 0$

$$(m-2)(m-1) = 0$$

$$\therefore m = 1, 2$$

$$C.F. = C_1 e^x + C_2 e^{2x}$$

$$\therefore y_1 = e^x, \; y_2 = e^{2x}$$

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2 \cdot e^{3x} - e^{3x} = e^{3x}.$$

Let, $P.I. = ue^x + ve^{2x}$ ——— (1)

Let, $u = \int \frac{-y_2 Q}{W} dx$

$$= -\int \frac{e^{2x} e^x}{e^{3x}} dx$$

$$= -\int \frac{e^{3x}}{e^{3x}} dx$$

$$= -\int dx$$

$$= -x$$

and $v = \int \frac{y_1 Q}{W} dx$

$$= \int \frac{e^x e^x}{e^{3x}} dx$$

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$= \int e^{-x}\, dx$

$= -e^{-x}$.

From (2), $P.I. = -xe^{x} - e^{2x}e^{-x}$

$= -xe^{x} - e^{x}$

$= -e^{x}(x+1)$

Hence, the G.S. is $y = c_1 e^{x} + c_2 e^{2x} - e^{x}(x+1)$.

(4) $(D^2 - 3D + 2)y = \dfrac{e^{x}}{1+e^{x}}$.

→

The A.E. is $m^2 - 2m + 2 = 0$

$(m-1)(m-2) = 0$

$\therefore m = 1, 2.$

$C.F. = c_1 e^{x} + c_2 e^{2x}$

$\therefore y_1 = e^{x}, \; y_2 = e^{2x}$

$$W = \begin{vmatrix} e^{x} & e^{2x} \\ e^{x} & 2e^{2x} \end{vmatrix} = e^{x}\cdot 2e^{2x} - e^{x}e^{2x} = 2e^{3x} - e^{3x} = e^{3x}$$

Let, $P.I. = u \cdot e^{x} + v \cdot e^{2x}$

where $u = \int -\dfrac{y_2 Q}{W}\, dx$

$= \int -\dfrac{e^{2x}\, e^{x}}{e^{3x}(1+e^{x})}$

$= -\int \dfrac{1}{1+e^{x}}\, dx$

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$$= -\int \frac{e^{-x}}{e^{-x}+1}\,dx$$ (Multiply & divide by e^{-x})

$$\int \frac{f'(x)}{f(x)}\,dx = \log f(x)$$

$$= \log(1+e^{-x})$$

and $$v = \int \frac{y_1 Q}{W}\,dx$$

$$= \int \frac{e^{x}\,e^{x}}{e^{3x}(1+e^{x})}\,dx$$

$$= \int \frac{e^{2x}}{e^{3x}(1+e^{x})}\,dx$$

$$= \int \frac{e^{-x}}{1+e^{x}}$$

$$= \int \frac{e^{-x}\,e^{-x}}{e^{-x}+1}\,dx$$

$$= \int \frac{e^{-x}\left[(e^{-x}+1)-1\right]}{e^{-x}+1}\,dx$$

$$= \int \left(e^{-x} - \frac{e^{-x}}{1+e^{-x}}\right)dx$$

$$= \int e^{-x}\,dx - \int \frac{e^{-x}}{1+e^{-x}}\,dx$$

$$= -e^{-x} + \log(1+e^{-x}) \quad \left[\int \frac{f(x)}{f(x)}\,dx = \log f(x)\right].$$

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⑤ Solve $(D^2+1)y = x\sin x$.

→

The A.E. is $m^2+1=0$

$m = \pm i$

$C.F. = C_1\cos x + C_2\sin x$.

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1$$

Let, $P.I = u\cdot\cos x + v\cdot\sin x$

where $u = \int \frac{-y_2 Q}{W}\,dx$

$$= -\int \frac{\sin x\,(x\sin x)}{1}\,dx$$

$$= -\int x\sin^2 x\,dx$$

$$= -\int x\left(\frac{1-\cos 2x}{2}\right)dx$$

$$= -\frac{1}{2}\int x\,dx + \frac{1}{2}\int x\cos 2x\,dx$$

$$= -\frac{x^2}{4} + \frac{1}{2}\left[x\int\cos 2x - \int\left[\frac{dx}{dx}\int\cos 2x\,dx\right]dx\right]$$

$$= -\frac{x^2}{4} + \frac{1}{2}\left[\frac{x\sin 2x}{2} - \int\left[+\frac{\sin 2x}{2}\right]dx\right]$$

$$= -\frac{x^2}{4} + \frac{1}{2}\left[\frac{x\sin 2x}{2} + \frac{\cos 2x}{4}\right]$$

$$= -\frac{x^2}{4} + \frac{1}{8}(2x\sin 2x + \cos 2x)$$

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$$v = \int \frac{y_1 g}{W} dx.$$

$$= \int \frac{\cos x \cdot (x \sin x)}{1} dx$$

$$= \int \frac{x}{2} \sin 2x \, dx$$

$$= \frac{1}{2} \int x \sin 2x \, dx$$

$$= \frac{1}{2} \left[x \int \sin 2x \, dx - \int \left[\frac{dx}{dx} \int \sin 2x \, dx \right] dx \right]$$

$$= \frac{1}{2} \left[x \left(\frac{-\cos 2x}{2} \right) - \int \left(\frac{-\cos 2x}{2} \right) dx \right]$$

$$= \frac{1}{2} \left[\frac{-x \cos 2x}{2} + \frac{1}{2} \frac{\sin 2x}{2} \right]$$

$$= \frac{1}{8} (-2x \cos 2x + \sin 2x).$$

$$\therefore P.I. = \left[\frac{-x^2}{4} + \frac{1}{8} (2x \sin 2x + \cos 2x \right] \cos x +$$

$$\left[\frac{1}{8} (-2x \cos 2x + \sin 2x) \right] \sin x$$

$$= -\frac{x^2}{4} \cos x + \frac{1}{8} \left[2x (\sin 2x \cos x - \sin x \cos 2x) \right.$$

$$\left. + (\cos 2x \cdot \cos x + \sin 2x \cdot \sin x) \right]$$

$$= -\frac{x^2}{4} \cos x + \frac{1}{8} \left[2x \sin (2x - x) + \cos (2x - x) \right]$$

$$= -\frac{x^2}{4} \cos x + \frac{1}{8} (2x \sin x + \cos x).$$

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- Hence, the G.S. is

$$y = C_1 \cos x + C_2 \sin x - \frac{x^2}{4}\cos x + \frac{1}{8}(2x \sin x + \cos x)$$

(6) Solve $(D^2+1)y = \operatorname{cosec} x \cdot \cot x$.

→

The A.E. is $m^2+1=0$

$$m = \pm i$$

$$C.F. = C_1 \cos x + C_2 \sin x$$

$$y_1 = \cos x, \quad y_2 = \sin x.$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1$$

Let, $P.I. = u \cos x + v \sin x$ ——— (1)

where, $u = \int \frac{-y_2 Q}{W}\, dx$

$$= \int \frac{-\sin x \cdot \operatorname{cosec} x \cdot \cot x}{1}\, dx$$

$$= -\int \cot x\, dx$$

$$= -\log(\sin x).$$

and $v = \int \frac{y_1 Q}{W}\, dx$

$$= \int \frac{\cos x \cdot \operatorname{cosec} x \cdot \cot x}{1}\, dx$$

$$= \int \cot^2 x\, dx$$

$$= \int (\operatorname{cosec}^2 x - 1)\, dx$$

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$\therefore v = -\cot x - x$

From (1),

P.I. $= -\log(\sin x)\cos x + (-\cot x - x)\sin x$

$= -\cos x \cdot \log(\sin x) - (\cot x + x)\sin x.$

$\therefore$ The G.S. is $y = c_1 \cos x + c_2 \sin x - \cos x \cdot \log(\sin x) - (\cot x + x)\sin x.$

(7) Solve $(D^2-1)y = e^{-x}\sin(e^{-x}) + \cos(e^{-x})$.

→

The A.E. is $m^2 - 1 = 0$

$(m+1)(m-1) = 0$

$\therefore m = -1, 1$

$\therefore$ C.F. $= c_1 e^x + c_2 e^{-x}$

$y_1 = e^x, \quad y_2 = e^{-x}.$

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -e^0 - e^0 = -2$$

Let, P.I. $= uy_1 + vy_2$

$= u \cdot e^x + v \cdot e^{-x}$ ——— (1)

where,

$$u = \int \frac{-y_2 Q}{W}\, dx$$

$$= -\int \frac{e^{-x}}{-2}\left[e^{-x}\sin(e^{-x}) + \cos(e^{-x})\right] dx$$

Let, $e^{-x} = t$

$\therefore -e^{-x}\, dx = dt.$

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$$\therefore u = \frac{-1}{2}\int (t\sin t + \cos t)\, dt$$

$$= -\frac{1}{2}\left[\int t\sin t\, dt + \int \cos t\, dt\right]$$

$$= -\frac{1}{2}\left[t\int \sin t\, dt - \int\left\{\frac{dt}{dt}\int \sin t\, dt\right\} dt\right] + \sin t$$

$$= -\frac{1}{2}\left[t(-\cos t) - \int (-\cos t)\, dt + \sin t\right]$$

$$= -\frac{1}{2}\left[-t\cos t + \sin t + \sin t\right]$$

$$= -\frac{1}{2}\left[-t\cos t + 2\sin t\right]$$

$$= \frac{1}{2}t\cos t - \sin t$$

$$= \frac{1}{2}e^{-x}\cos(e^{-x}) - \sin(e^{-x})$$

and $v = \int \frac{y_1 Q}{W}\, dx$

$$= \int \frac{e^{2x}}{-2}\left[e^{-x}\sin(e^{-x}) + \cos(e^{-x})\right] dx$$

$$= -\frac{1}{2}\int \left[\sin(e^{-x})\right]$$

$$= \int \frac{e^{2x}\left[\cos(e^{-x}) + e^{-x}\sin(e^{-x})\right]}{-2} dx$$

$$= -\frac{1}{2}e^{x}\cos(e^{-x}) \qquad \left[\int e^{x}\{f(x) + f'(x)\}\, dx = e^{x} f(x).\right]$$

$$= -\frac{1}{2}e^{x}\cos e^{x}$$

From ①,

$$\text{P.I.} = \frac{1}{2}e^{-x}\cos(e^{-x})\cdot e^{x} - \frac{1}{2}e^{-x}\cos(e^{-x})\, e^{-x} - \sin(e^{-x})\, e^{x}$$

$$= -e^{x}\sin(e^{-x}).$$

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- The G.S. is $y = c_1 e^{x} + c_2 e^{-x} - e^{x} \sin(e^{-x})$.

(8) Solve $(D^2+5D+6)\,y = e^{-2x}\sec^2 x\,(1+2\tan x)$.

→

The A.E. is $m^2+5m+6=0$

$(m+2)(m+3)=0$

$m = -2, -3$

$C.F. = c_1 e^{-2x} + c_2 e^{-3x}$.

$y_1 = e^{-2x}$, $y_2 = e^{-3x}$

$W = -e^{-5x}$

$$= \begin{vmatrix} e^{-2x} & e^{-3x} \\ -2e^{-2x} & -3e^{-3x} \end{vmatrix} = -3e^{-5x} + 2e^{-5x} = -e^{-5x}$$

Let, $P.I. = u\cdot y_1 + v y_2$ ——— (1)

where $u = \int \frac{-y_2 Q}{W}\,dx$

$$= \int \frac{-e^{-3x}\, e^{-2x} \sec^2 x\,(1+2\tan x)}{-e^{-5x}}\,dx$$

$$= \int \sec^2 x\,(1+2\tan x)\,dx \qquad \left[\int f(x)\cdot f'(x)\,dx = \frac{\{f(x)\}^2}{2}\right]$$

$$= \int \frac{2\sec^2 x\,(1+2\tan x)}{2}\,dx$$

$f(x) = 1+2\tan x$

$$= \frac{1}{2}\int 2(1+\tan^2 x)$$

$$= \frac{1}{2}\,\frac{(1+2\tan x)^2}{2}$$

$$= \frac{1}{4}(1+2\tan x)^2.$$

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$$v = \int \frac{y_1 Q}{W} dx$$

$$= \int \frac{e^{-2x} \cdot e^{-2x} \sec^2 x (1+2\tan x)}{-e^{-5x}} dx$$

$$= -\int e^{x} \sec^2 x (1+2\tan x) dx$$

$$= -\int e^{x} (\sec^2 x + 2\sec^2 x \tan x) dx$$

$$= -e^{x} \sec^2 x \quad \left[\because \int e^{x} \{f(x) + f'(x)\} dx = e^{x} f(x) \right.$$
$$\left. \text{Here, } f(x) = \sec^2 x \right]$$

From (1),

$$P.I. = \frac{1}{4} (1+2\tan x)^2 e^{-2x} - e^{x} \sec^2 x \, e^{-3x}$$

$$= \frac{1}{4} (1+2\tan x)^2 e^{-2x} - e^{-2x} \sec^2 x.$$

Hence, the G.S. is

$$y = C_1 e^{-2x} + C_2 e^{-3x} + \frac{1}{4} (1+2\tan x)^2 e^{-2x} - e^{-2x} \sec^2 x$$

(9) Solve $(D^3 - 6D^2 + 12D - 8) y = \frac{e^{2x}}{x}$.

→ The A.E. is $m^3 - 6m^2 + 12m - 8 = 0$

$$(m-2)^3 = 0$$

$$\therefore m = 2, 2, 2.$$

$$\therefore C.F. = e^{2x} (C_1 + C_2 x + C_3 x^2).$$

$$= C_1 e^{2x} + C_2 x e^{2x} + C_3 x^2 e^{2x}.$$

$$\therefore y_1 = e^{2x}, \; y_2 = x e^{2x}, \; y_3 = x^2 e^{2x}.$$

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$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

$$= \begin{vmatrix} e^{2x} & xe^{2x} & x^2e^{2x} \\ 2e^{2x} & e^{2x}+2xe^{2x} & 2xe^{2x}+2x^2e^{2x} \\ 4e^{2x} & 2e^{2x}+2e^{2x}+4xe^{2x} & 2e^{2x}+4xe^{2x}+4xe^{2x}+4x^2e^{2x} \end{vmatrix}$$

$$= \begin{vmatrix} e^{2x} & xe^{2x} & x^2e^{2x} \\ 2e^{2x} & e^{2x}+2xe^{2x} & 2xe^{2x}+2x^2e^{2x} \\ 4e^{2x} & 4e^{2x}+4xe^{2x} & 2e^{2x}+8xe^{2x}+4x^2e^{2x} \end{vmatrix}$$

$$= e^{6x} \begin{vmatrix} 1 & x & x^2 \\ 2 & 1+2x & 2x+2x^2 \\ 4 & 4+4x & 2+8x+4x^2 \end{vmatrix}$$

$$= e^{6x}\left[1\{(1+2x)(2+8x+4x^2)-(2x+2x^2)(4+4x)\} - x\{2(2+8x+4x^2)-4(2x+2x^2)\}+x^2\{2(2+8x+4x^2)-4(2x+2x^2)\}\right]$$

$$= e^{6x}\left[2+8x+4x^2+4x+16x^2+8x^3-\{8x+8x^2+8x^2+8x^3\}-x\{4+16x+8x^2-8x-8x^2\}+x^2\{4+16x+8x^2-8x-8x^2\}\right]$$

$$= e^{6x}\left[8x^3+20x^2+12x+2-8x-8x^2-8x^2-8x^3-4x-16x^2-8x^3+8x^2+8x^3+4x^2+16x^3+8x^3-8x^3-8x^3\right]$$

$$= e^{6x}\left[8x^3 \quad 2\right]$$

$$= 2e^{6x}$$

Let, P.I. $= uy_1 + vy_2 + wy_3$ ——— (1)

$$= u \cdot e^{2x} + v\,xe^{2x} + w \cdot x^2e^{2x}$$ ——— (1)

Let, $u = \int \frac{(y_2y_3' - y_3y_2')Q}{W}\,dx$

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$$u = \int \frac{[xe^{2x}(2xe^{2x} + 2x^2e^{2x}) - x^2e^{2x}(e^{2x} + 2xe^{2x})}{2e^{6x}}$$

$$= \int \frac{(2x^2e^{4x} + 2x^3e^{4x} - x^2e^{4x} - 2x^3e^{4x})}{2e^{6x}} \frac{e^{2x}}{x} dx$$

$$= \int \frac{x^2e^{4x}}{2e^{6x}} \frac{e^{2x}}{x} dx$$

$$= \int \frac{x}{2} dx$$

$$= \frac{x^2}{4}$$

$$v = \int \frac{(y_3y_1' - y_1y_3')}{W} Q \, dx$$

$$= \int \frac{(x^2e^{2x} \cdot 2e^{2x} - e^{2x}\{2xe^{2x} + 2x^2e^{2x}\})}{2e^{6x}} \frac{e^{2x}}{x} dx$$

$$= \int \frac{(2x^2e^{4x} - 2xe^{4x} - 2x^2e^{4x})}{2e^{6x}} \frac{e^{2x}}{x} dx$$

$$= \int \frac{(-2xe^{4x})}{2e^{6x}} \frac{e^{2x}}{x} dx$$

$$= \int -dx$$

$$= -x.$$

$$w = \int \frac{(y_1y_2' - y_2y_1')}{W} Q \, dx$$

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$$W = \int \frac{\left[e^{2x}(e^{2x} + 2xe^{2x}) - xe^{2x}\cdot 2e^{2x}\right]}{2e^{6x}} \frac{e^{2x}}{x}\,dx$$

$$= \int \frac{(e^{4x} + 2xe^{4x} - 2xe^{4x})}{2e^{6x}} \frac{e^{2x}}{x}\,dx$$

$$= \int \frac{e^{4x}\cdot e^{2x}}{2e^{6x}\,x}\,dx$$

$$= \int \frac{1}{2x}\,dx$$

$$= \frac{1}{2}\log x.$$

From (1),

$$P.I. = \frac{x^2}{4}e^{2x} + (-x)\,xe^{2x} + \left(\frac{1}{2}\log x\right)x^2e^{2x}$$

$$= \frac{x^2}{4}e^{2x} - x^2e^{2x} + \left(\frac{1}{2}\log x\right)x^2e^{2x}$$

$$= -\frac{3}{4}x^2e^{2x} + \frac{x^2}{2}e^{2x}\log x.$$

Hence, the G.S. is

$$y = (C_1 + C_2x + C_3x^2)e^{2x} - \frac{3}{4}x^2e^{2x} + \frac{x^2}{2}e^{2x}\log x.$$

* Euler - Cauchy (Linear) Equations :-

An eq^n of the form

$$a_0x^n\frac{d^ny}{dx^n} + a_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + a_2x^{n-2}\frac{d^{n-2}y}{dx^{n-2}} + \cdots + a_{n-1}x\frac{dy}{dx} + a_ny = Q(x) \quad \text{——(1)}$$

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